Learning Illumination to Reduce Structural Ambiguity in 3D Fluorescence Microscopy

Supplementary Material

This supplement gives the details of the optical study (Secs. S1 to S4), the reduced-field structural analysis (Secs. S5 and S9 to S14), a preliminary single-stack study (Sec. S15) and implementation details (Sec. S16). Section, figure and table numbers of the main paper are written without the prefix S.

S1. Optical encoders and photon model

Encoders. All four encoders share the NA-0.8 objective pupil, the vectorial Richards–Wolf excitation relay, the $0.32\,\mu\mathrm{m}$ analysis binning, the NA-0.1 emission PSF, the 12×12 camera, 256 frames and the voxel-integrated assembly; only the field reaching the relay object plane differs. *Phase plate*: tilted plane wave times a fixed random phase plate ($0.32\,\mu\mathrm{m}$ features, phases uniform on $[0, 2\pi)$) propagated over $D = 6.6\,\mu\mathrm{m}$. *No plate*: the same angle scan and propagation without the plate. *In-focus DMD*: 256 random binary amplitude patterns ($0.32\,\mu\mathrm{m}$ pixels, 50% on) imaged in focus. *Amplitude mask*: the angle scan through a fixed random binary amplitude mask ($0.32\,\mu\mathrm{m}$, 50% open) at the plate's position.

Illuminated aperture. If the simulated illuminated aperture equals the $30.7\,\mu\mathrm{m}$ field, its hard edge, $6.6\,\mu\mathrm{m}$ before the image plane, is itself a defocused amplitude encoder: tilted-beam edges sweep laterally by $z \tan\theta$ and their diffraction ripples reach the field centre, so even the no-plate control acquires depth structure. The comparison of the main paper therefore simulates every encoder on an optical grid twice the field width ($61.4\,\mu\mathrm{m}$ aperture) and crops the excitation to the field, as in a microscope whose illuminated area exceeds its field of view. This geometry was adopted after a first comparison with the field-sized aperture revealed the effect; that comparison is reported in Sec. S3. Twice the field is not yet enough for the no-plate control: per-pixel excitation sequences in the central third of the field still vary by $5.6\,\%$ between directions after removing each frame's mean brightness ($37\,\%$ at the field border), against $70\,\%$ for the plate. At four times the field ($123\,\mu\mathrm{m}$ aperture) this falls to $1.0\,\%$ ($1.2\,\%$ at the border) and the no-plate PnP depth error rises from 0.38, 0.50, 0.52 $\mu\mathrm{m}$ to 0.75, 1.04, 1.10 $\mu\mathrm{m}$ at $C = 1, 4, 8$ (PSNR 30.9, 30.4, 30.2 dB; TV: 0.30, 0.30, 0.34 $\mu\mathrm{m}$ to 0.48, 0.71, 0.74 $\mu\mathrm{m}$), while the plate changes by at most 0.07 dB and $0.02\,\mu\mathrm{m}$ (parameters re-tuned on the extended grid; no edge selections). The main table therefore reports the no-plate row at four times the field; the remaining $1\,\%$ modulation, decodable only with many frames, is why its depth error stays below the depth-blind oracle at $C = 1$.

encoder	full field	reduced field	build (s)	incident power
phase plate	6.2e-07	1.4e-07	327	–
no plate	6.1e-07	1.3e-07	43	–
in-focus DMD	6.0e-07	1.3e-07	42	–
amplitude mask	6.3e-07	1.3e-07	36	–

encoder	full field	reduced field	build (s)	incident power
phase plate	8.0e-07	1.4e-07	111	0.277
no plate	1.4e-06	1.3e-07	186	0.570
in-focus DMD	6.3e-07	1.3e-07	104	0.378
amplitude mask	9.6e-07	1.4e-07	111	0.211

Table S1. Full-size operators ($36{,}864^2$) for the overfilled (top) and the field-sized aperture (bottom): largest relative ℓ_2 difference from the live forward model on three random volumes (full and reduced field; tolerance 10^{-5}), assembly time on one GPU, and the fraction of incident power that passes the pupil.

Photon model. Each encoder's excitation is normalised to the same mean over all frames and the analysis volume (equal deposited excitation dose). Counts are $\mathrm{Poisson}(\kappa_{\mathrm{frame}} y_i)$ with $\kappa_{\mathrm{frame}} = \kappa_0\, 256/k$. The exposure κ_0 of each specimen brings the plate to 1000 photons per pixel per frame at the mean signal at $k = 256$ (15 dB); it is computed from the forward model alone. At equal dose the mean detected signal of every encoder is within 4% of the plate's on every test crop with the overfilled aperture (within 17% with the field-sized aperture). Incident power through the pupil (Tab. S1) sets the laser power required, not the specimen dose: with the field-sized aperture the plate passes 28% and the amplitude mask 21% of the incident light.

Specimens and tuning. Test: two crops from each of six cells of three animals (DEEP^2, public under the MIT licence [4]), selected from the reference volumes alone and frozen before any reconstruction. Validation: two crops of a separate vascular volume (DnCNN prior, TV) and a held-out bead layout (sparsity prior). The penalty scale ρ and the L1 scale are selected per encoder and compression by mean validation PSNR (Tab. S2); Sec. S4 repeats the selection with an expanded grid.

encoder	prior	$C = 1$	$C = 4$	$C = 8$
phase plate	DnCNN	2/0	0.5/0	0.25/0
	sparsity	1/4	0.5/4	0.25/4
no plate	DnCNN	0.5/0	0.125/0	0.125/0
	sparsity	0.5/4	0.25/4	0.25/4
in-focus DMD	DnCNN	1/0.0625	0.25/0	0.125/0
	sparsity	1/4	0.25/4	0.5/4
amplitude mask	DnCNN	2/0	0.5/0	0.5/0
	sparsity	1/4	0.5/4	0.5/4

Table S2. Selected PnP parameters (ρ scale / L1 scale) on the initial grid, overfilled aperture. Selections at $C = 4, 8$ on the expanded grid are in Tab. S11.

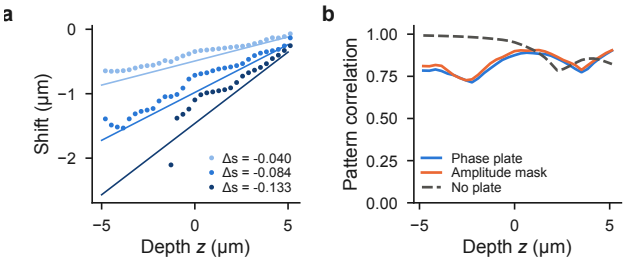


Figure S1. (a) Lateral shift of the plate's excitation between scan pattern 0 and patterns 1–3 (same azimuth) versus depth; lines: thin-screen prediction pivoting at the plate's conjugate plane. (b) Peak normalised correlation between scan patterns 0 and 1 per depth.

S2. Mechanism and point-emitter diagnostics

Depth structure of the excitation. Table S3 summarises, in the central half-field, how the per-pixel temporal excitation signature (its 256 values) decorrelates with depth. Without the plate every depth receives essentially the same sequence. The in-focus DMD decorrelates with depth as fast as the plate but is identical at mirror depths. The plate decorrelates within about $1\,\mu$m and its mirror depths are uncorrelated.

Shear with input tilt. Figure S1 measures the lateral translation of the plate's excitation between scan angles at every depth. The shift is linear in depth, with slopes 0.74–0.94 of the thin-screen prediction $\Delta s/\sqrt{1-s^2}$ and a pivot 6.2–7.0 µm from focus, against the plate's conjugate plane at $D = 6.6\,\mu$m. Patterns also partly decorrelate between angles because the pupil passes a different part of the tilted angular spectrum. The amplitude mask's patterns decorrelate between angles like the plate's, whereas the no-plate excitation barely changes.

Point-emitter diagnostics. Table S4 uses operator columns for 25 interior lateral positions and 16 depths, with a Gaussian approximation of shot noise at the plate's 15 dB exposure and unknown brightness. These quantities describe the information in the measurements, not optical resolution.

encoder	contrast (focus)	signature corr. at Δz (µm) 0.32	0.64	1.28	2.56	mirror corr.	border contrast
phase plate	0.71	0.86	0.53	0.09	0.05	0.05	0.71
no plate	0.18	1.00	1.00	1.00	1.00	1.00	0.25
in-focus DMD	0.82	0.96	0.56	0.13	0.02	1.00	0.82
defocused amplitude mask	0.54	0.92	0.72	0.44	0.43	0.44	0.55

encoder	contrast (focus)	signature corr. at Δz (µm) 0.32	0.64	1.28	2.56	mirror corr.	border contrast
phase plate	0.72	0.86	0.54	0.10	0.05	0.05	0.83
no plate	0.38	1.00	1.00	0.98	0.94	0.96	0.81
in-focus DMD	0.82	0.96	0.56	0.12	0.02	1.00	0.82
defocused amplitude mask	0.54	0.92	0.73	0.44	0.41	0.43	0.79

Table S3. Excitation statistics in the central half-field for the overfilled (left) and the field-sized aperture (right): lateral contrast at focus, median correlation of per-pixel temporal signatures Δz apart and at mirror depths about focus, and the lateral contrast at the field border.

encoder	k	cos(adjacent)	cos(mirror)	axial CRB (µm)	mirror d'	wrong depth (%)	mirror (%)
phase plate	256	0.940	0.823	0.134	10.2	1.0	0.01
	32	0.943	0.828	0.137	10.2	1.6	0.04
no plate	256	1.000	0.999	3.373	0.8	88.2	4.10
	32	1.000	0.999	3.376	0.7	88.7	4.25
in-focus DMD	256	0.967	0.990	0.183	2.4	24.6	13.23
	32	0.970	0.991	0.185	2.4	25.1	13.21
defocused amplitude mask	256	0.966	0.904	0.171	8.0	3.1	0.06
	32	0.969	0.908	0.180	7.7	4.5	0.09

encoder	k	cos(adjacent)	cos(mirror)	axial CRB (µm)	mirror d'	wrong depth (%)	mirror (%)
phase plate	256	0.941	0.807	0.113	12.4	0.9	0.01
	32	0.944	0.810	0.116	12.3	1.2	0.02
no plate	256	0.995	0.925	0.327	9.2	34.8	0.36
	32	0.996	0.915	0.361	9.1	36.8	0.37
in-focus DMD	256	0.968	0.991	0.180	2.4	24.0	12.97
	32	0.970	0.991	0.183	2.4	24.5	12.91
defocused amplitude mask	256	0.965	0.861	0.131	11.4	1.4	0.00
	32	0.968	0.860	0.137	11.4	2.2	0.02

Table S4. Point-emitter diagnostics, overfilled (top) and field-sized aperture (bottom): median cosine between operator columns of depth-adjacent ($0.64\,\mu$m) and mirror voxels, axial Cramér–Rao bound, mirror separation d', and Monte Carlo maximum-likelihood depth errors (any wrong depth / the mirror depth).

S3. Comparison with a field-sized aperture

With the field-sized aperture (Tab. S5 and Fig. S2) the aperture edge gives the no-plate scan and the amplitude mask depth structure: they match or exceed the plate in depth error, while the in-focus DMD result is unchanged. Equalising detected photons instead of deposited dose (Tab. S6) changes the plate and no-plate results by at most 0.1 dB.

C	encoder	PSNR	FG-SSIM	depth (μm)	bead (μm)	ghost
1	phase plate	34.25	0.778	0.329	0.00	0.035
	no plate	34.22 (4/12)	0.763 (7/12)	0.254 (1/12)	0.00	0.024
	in-focus DMD	28.76 (12/12)	0.485 (12/12)	2.169 (12/12)	2.99	1.007
	amp. mask	34.92 (2/12)	0.805 (1/12)	0.230 (0/12)	0.00	0.026
4	phase plate	33.23	0.712	0.424	0.00	0.042
	no plate	33.81 (2/12)	0.744 (3/12)	0.279 (0/12)	0.02	0.015
	in-focus DMD	28.70 (11/12)	0.476 (11/12)	2.170 (12/12)	3.29	1.010
	amp. mask	34.39 (0/12)	0.785 (0/12)	0.263 (0/12)	0.00	0.038
8	phase plate	32.87	0.698	0.466	0.00	0.062
	no plate	33.03 (4/12)	0.703 (6/12)	0.323 (1/12)	0.04	0.019
	in-focus DMD	28.65 (11/12)	0.468 (11/12)	2.170 (12/12)	3.14	1.015
	amp. mask	33.40 (0/12)	0.728 (0/12)	0.317 (0/12)	0.04	0.039

Table S5. Encoder comparison with the field-sized aperture (PnP, initial grid). In parentheses: crops on which the plate is better.

C	encoder	PSNR	FG-SSIM	depth (μm)	bead (μm)	ghost
1	phase plate	34.18	0.776	0.327	0.00	0.035
	no plate	34.22 (2/12)	0.762 (7/12)	0.253 (1/12)	0.00	0.023
	in-focus DMD	28.87 (5/5)	0.493 (5/5)	2.097 (5/5)	–	–
8	phase plate	32.90	0.698	0.467	0.00	0.063
	no plate	33.02 (4/12)	0.702 (6/12)	0.322 (1/12)	0.06	0.020

Table S6. Equal detected photons instead of equal dose (field-sized aperture, parameters of Tab. S5). The DMD was evaluated on 5 of 12 crops at $C = 1$ and the amplitude mask was not evaluated in this setting.

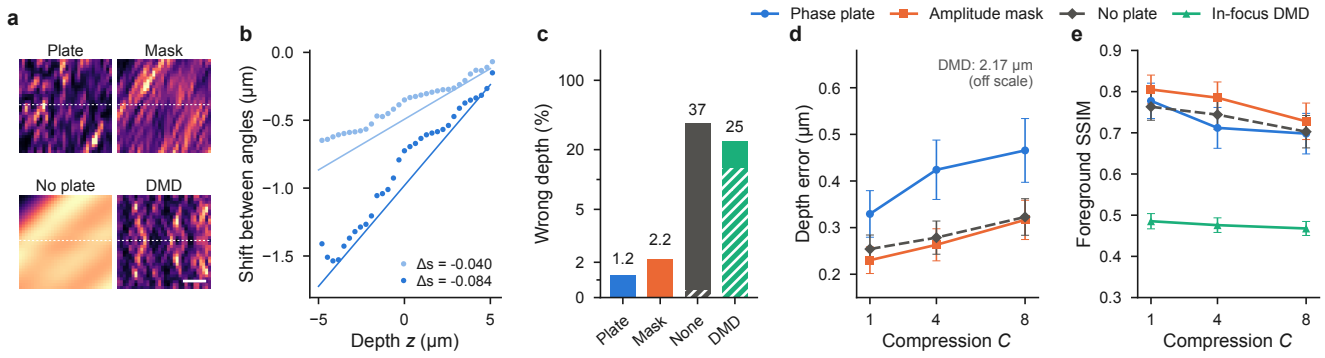


Figure S2. Main-paper Fig. 2 for the field-sized aperture (one realisation, PnP): (a) excitation xz sections, (b) shear with tilt, (c) point emitters assigned a wrong depth at $k = 32$, (d, e) depth error and foreground SSIM versus compression (mean $\pm$ s.e.m. over 12 crops).

S4. Robustness of the encoder comparison

Protocol. Before any result of this study was seen we fixed: three realisations of the phase plate and of the amplitude mask (seeds 0, 101, 202; realisation r draws the central encoder with seed r and the surround of the overfilled aperture with seed $r + 1$, so $r = 0$ is the encoder of the main comparison); a defocus sweep $D \in \{3.3, 6.6, 13.2\}$ μm for plate realisation 0; compressions $C = 8$ and 4; and an expanded tuning grid for every encoder (DnCNN prior: ρ scale $\in \{2^{-4}, \ldots, 4\} \times$ L1 scale $\in \{0, 0.0625\}$; sparsity prior: ρ scale $\in \{0.125, \ldots, 2\} \times$ L1 scale $\in \{1, 4, 16\}$), extended automatically by halving or doubling ρ (or multiplying the L1 scale by 4) whenever the optimum lay on an edge, at most twice. Every new operator was checked against the live forward model (relative error $\leq 6.3 \times 10^{-7}$). Relay, detector, dose, photon model, noise seeds, direction subsets, objects and metrics are those of the main comparison. Paired differences are computed per crop (mean over five direction subsets), with cell-level bootstrap intervals (6 cells, 10^4 resamples); realisations are summarised separately and never

pooled as specimens.

Defocus. Halving D degrades every metric at $C = 8$ (PSNR worse on 12 of 12 crops) and less at $C = 4$; doubling it leaves PSNR unchanged within the interval and lowers depth error slightly (0.016–$0.018\,\mu$m, 9 of 12 crops; Tab. S10). Single-pattern statistics barely change with D (lateral contrast 0.71, depth correlation 0.53 at $0.64\,\mu$m, mirror correlation 0.05), whereas sequence-level quantities do: the median cosine between the $k = 32$ sequences of adjacent depths falls from 0.948 to 0.943, and between mirror depths from 0.854 to 0.823, as D grows, and the shear pivot follows D. At $D = 13.2\,\mu$m only the smallest scan step gives a reliable fit; for larger steps the pattern decorrelates within the correlation window.

Tuning. No final selection lies on an edge of the extended grid (Tab. S11). For neurons only the no-plate scan at $C = 8$ changed: its test PSNR rose by 0.43 dB and its depth error fell by $0.085\,\mu$m. The sparsity prior for beads moved from L1 scale 4 to 16 for most encoders, which changes bead depth errors by at most $0.02\,\mu$m for the defocused encoders and $0.17\,\mu$m for the DMD (whose beads lie at mirror depths in both cases).

Total-variation baseline. We solve $\arg\min_{0\le\mathbf{x}\le1} \frac{1}{2}\|\mathbf{A}\mathbf{x} - \mathbf{y}\|_2^2 + \lambda\,\mathrm{TV}(\mathbf{x})$, isotropic 3D TV of forward differences, by ADMM [2] (penalties 1, conjugate-gradient x-update with at most 20 iterations at tolerance 10^{-5}, 300 iterations) on the same operators and noisy measurements. $\lambda \in \{0.01, 0.03, 0.1, 0.3, 1, 3\}$ is tuned per encoder and compression on the vascular validation crops with the same edge-extension rule. The relative change of $\mathbf{x}$ over the last 10 iterations has medians of 1.3×10^{-3}–1.9×10^{-2} per encoder and compression (largest at $C = 1$), so most solves do not meet a 10^{-3} criterion and we do not claim exact minimisers. Re-solving 12 prespecified test problems at $C = 8$ with 1000 iterations changed PSNR by -0.03 dB on average (at most 0.18 dB), foreground SSIM by at most 0.002 and depth error by at most $0.001\,\mu$m; four problems at $C = 1$ changed by at most 0.03 dB and $0.006\,\mu$m.

Qualitative comparison. Figures S5 and S6 show the median test crop and the bead layout at $C = 8$ for all encoders. Without the plate, depth is largely lost; the DMD places structure at mirror depths.

C	encoder	seed	PSNR	FG-SSIM	depth (μm)	bead (μm)	ghost
8	phase plate	0	34.81	0.784	0.431	0.00	0.051
8	phase plate	101	34.93	0.787	0.422	0.02	0.037
8	phase plate	202	34.85	0.782	0.438	0.02	0.029
8	amp. mask	0	33.70	0.703	0.523	0.06	0.044
8	amp. mask	101	34.37	0.766	0.419	0.17	0.067
8	amp. mask	202	34.36	0.763	0.396	0.00	0.035
4	phase plate	0	35.36	0.801	0.384	0.02	0.014
4	phase plate	101	35.38	0.801	0.391	0.00	0.015
4	phase plate	202	35.45	0.800	0.392	0.00	0.012
4	amp. mask	0	34.83	0.773	0.392	0.00	0.020
4	amp. mask	101	34.95	0.782	0.382	0.02	0.019
4	amp. mask	202	34.87	0.779	0.370	0.00	0.014

Table S7. Per-realisation results (multi-slice PnP, 12 test crops; beads: depth error of detected beads and mirror-ghost ratio).

C	seed	ΔPSNR (dB)	ΔFG-SSIM	Δdepth (μm)	cells (PSNR)
8	0	+1.10 [+0.71, +1.53] (12/12)	+0.081 [+0.062, +0.108] (12/12)	+0.091 [+0.067, +0.122] (11/12)	6/6
8	101	+0.56 [+0.28, +0.99] (11/12)	+0.021 [+0.009, +0.038] (12/12)	-0.003 [-0.062, +0.063] (4/12)	6/6
8	202	+0.48 [+0.23, +0.78] (10/12)	+0.019 [+0.008, +0.030] (11/12)	-0.042 [-0.096, +0.017] (4/12)	5/6
4	0	+0.53 [+0.25, +0.81] (11/12)	+0.028 [+0.019, +0.040] (12/12)	+0.008 [-0.018, +0.034] (7/12)	6/6
4	101	+0.43 [+0.19, +0.73] (12/12)	+0.018 [+0.011, +0.031] (12/12)	-0.009 [-0.051, +0.033] (5/12)	6/6
4	202	+0.58 [+0.30, +0.91] (12/12)	+0.021 [+0.014, +0.031] (12/12)	-0.022 [-0.059, +0.019] (4/12)	6/6

Table S8. Phase plate minus amplitude mask for matched realisations (positive: plate better; depth: reduction of depth error). Brackets: cell-bootstrap 95% interval; parentheses: crops favouring the plate; last column: cells favouring the plate in PSNR. Over all nine plate/mask pairings, including mismatched seeds, the plate is better in PSNR by 0.44–1.23 dB at $C = 8$ and 0.40–0.62 dB at $C = 4$ and in foreground SSIM by 0.016–0.084, while the depth-error difference ranges from -0.042 to $+0.101$ μm.

C	seed	SOM004 c1	SOM004 c2	SOM005 c1	SOM006 c1	SOM006 c2	SOM006 c3
8	0	+0.67	+1.96	+0.52	+0.81	+1.57	+1.09
8	101	+0.42	+0.39	+0.18	+0.23	+1.57	+0.59
8	202	+0.38	+0.45	+0.36	-0.02	+1.12	+0.61
4	0	+0.30	+1.07	+0.04	+0.43	+0.92	+0.41
4	101	+0.07	+0.54	+0.16	+0.31	+1.13	+0.35
4	202	+0.21	+0.79	+0.31	+0.15	+1.23	+0.79

Table S9. Phase plate minus amplitude mask PSNR (dB) per cell (mean over its two crops). 35 of 36 entries are positive. By animal, the plate's PnP PSNR at $C = 8$ spans 34.2–35.9 dB over realisations, with the same encoder ranking in every animal.

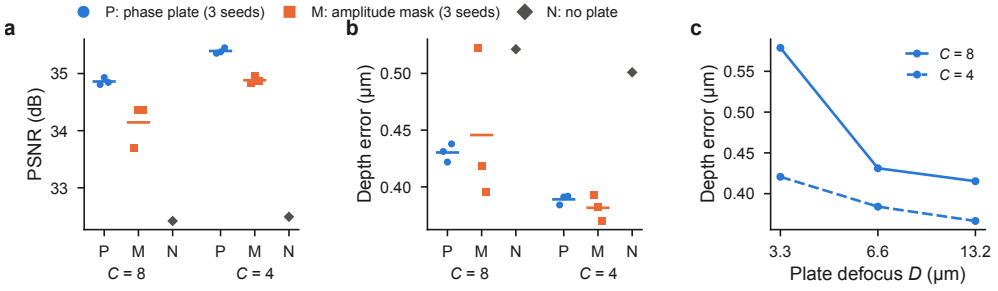


Figure S3. (a, b) PSNR and depth error of the three plate (P) and three amplitude-mask (M) realisations (markers; bar: mean) and of the no-plate scan (N). (c) Plate depth error versus defocus D.

C	D (μm)	PSNR	FG-SSIM	depth	bead	ΔPSNR vs. 6.6	wrong depth (%)	adj. cos.	mirror cos.	pivot (μm) / slope
8	3.3	33.80	0.709	0.579	0.06	-1.01 [-1.40, -0.62]	2.0	0.948	0.854	3.2 / 0.78
8	6.6	34.81	0.784	0.431	0.00	–	1.6	0.943	0.827	6.5 / 0.81
8	13.2	34.69	0.783	0.415	0.09	-0.11 [-0.30, +0.08]	1.5	0.943	0.823	15.3 / 0.66
4	3.3	34.97	0.785	0.421	0.02	-0.39 [-0.73, -0.09]	2.0	0.948	0.854	3.2 / 0.78
4	6.6	35.36	0.801	0.384	0.02	–	1.6	0.943	0.827	6.5 / 0.81
4	13.2	35.26	0.802	0.367	0.00	-0.09 [-0.23, +0.06]	1.5	0.943	0.823	15.3 / 0.66

Table S10. Plate defocus sweep (realisation 0, PnP). ΔPSNR: paired against $D = 6.6\,\mu$m with cell-bootstrap interval. Wrong depth, adjacent and mirror cosine: point-emitter diagnostics at $k = 32$. Pivot / slope ratio: shear fit between scan patterns 0 and 1 (thin-screen prediction: pivot D, ratio 1).

		neurons (DnCNN)		beads (L1)		neuron test	
encoder	C	original	expanded (ext.)	original	expanded (ext.)	PSNR	depth (μm)
phase plate	8	0.25/0	0.25/0 (0)	0.25/4	0.5/16 (1)	34.81 → 34.81	0.431 → 0.431
phase plate	4	0.5/0	0.5/0 (0)	0.5/4	1/16 (1)	35.36 → 35.36	0.384 → 0.384
no plate	8	0.125/0	0.0625/0 (1)	0.25/4	0.25/4 (0)	31.99 → 32.42	0.606 → 0.521
no plate	4	0.125/0	0.125/0 (0)	0.25/4	0.125/4 (1)	32.49 → 32.49	0.500 → 0.501
in-focus DMD	8	0.125/0	0.125/0 (0)	0.5/4	0.5/4 (0)	28.83 → 28.83	2.169 → 2.169
in-focus DMD	4	0.25/0	0.25/0 (0)	0.25/4	1/16 (1)	28.91 → 28.91	2.168 → 2.168
amp. mask	8	0.5/0	0.5/0 (0)	0.5/4	0.5/16 (1)	33.70 → 33.70	0.523 → 0.523
amp. mask	4	0.5/0	0.5/0 (0)	0.5/4	1/16 (1)	34.83 → 34.83	0.393 → 0.392

Table S11. Initial versus expanded tuning (ρ scale / L1 scale; in parentheses the number of automatic grid extensions) and the resulting neuron test means (initial → expanded).

C	encoder	λ	PSNR	FG-SSIM	depth (μm)	plate adv. PSNR	s	Δx
8	phase plate	0.1	37.00	0.847	0.243	–	9.5	1.3e-03
8	amp. mask	0.1	36.74	0.827	0.237	+0.26 [-0.04, +0.55] (9/12)	11.1	1.7e-03
8	no plate	0.03	34.34	0.676	0.342	+2.66 [+1.54, +3.78] (12/12)	9.9	2.6e-03
8	in-focus DMD	0.1	28.53	0.473	2.311	+8.47 [+6.74, +9.89] (12/12)	12.5	2.5e-03
4	phase plate	0.3	37.34	0.852	0.223	–	20.6	2.8e-03
4	amp. mask	0.3	36.79	0.823	0.239	+0.55 [+0.29, +0.82] (11/12)	18.9	3.8e-03
4	no plate	0.03	34.74	0.703	0.302	+2.60 [+1.22, +4.01] (12/12)	11.3	5.0e-03
4	in-focus DMD	0.3	28.49	0.465	2.326	+8.85 [+6.93, +10.42] (12/12)	17.5	6.4e-03
1	phase plate	1	38.15	0.881	0.187	–	55.3	9.1e-03
1	amp. mask	1	37.59	0.857	0.191	+0.56 [+0.22, +0.88] (10/12)	59.9	7.8e-03
1	no plate	0.3	34.39	0.682	0.295	+3.76 [+2.45, +5.21] (12/12)	55.3	1.8e-02
1	in-focus DMD	3	28.48	0.435	2.285	+9.67 [+7.82, +11.34] (12/12)	86.2	1.9e-02

Table S12. TV baseline (realisation 0; 12 test crops, five direction subsets for $C > 1$). Plate adv.: plate minus encoder PSNR with cell-bootstrap interval and crops favouring the plate; s: median solve time on one GPU; Δx: median relative change over the last 10 iterations.

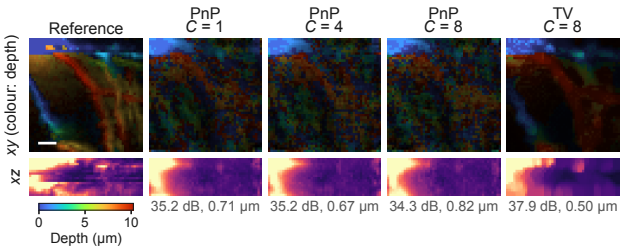


Figure S4. **Full-size recovery with the phase plate** on the prespecified display crop (largest reference depth spread), one direction subset. Top: lateral maximum-intensity projection, colour coding the depth of the brightest voxel; bottom: xz projection. Identical display scale for all panels; below: PSNR and local depth error of this crop. Bar: $5\,\mu$m.

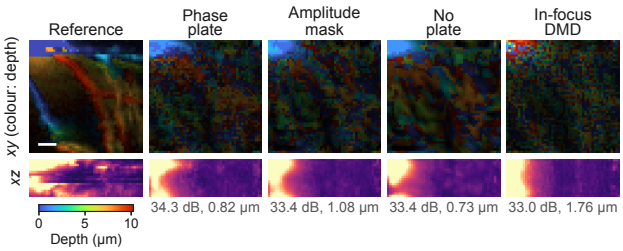


Figure S5. All encoders at $C = 8$ (PnP, one direction subset) on the crop of Fig. S4: depth-coded lateral projection and xz projection, identical display scale; below: PSNR and depth error of this crop.

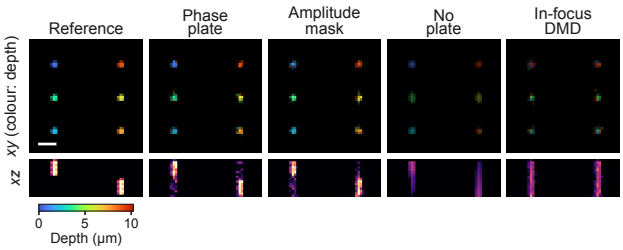


Figure S6. Bead layout at $C = 8$ (PnP, one direction subset) for all encoders, as in Fig. S5.

S5. Reduced-field structural outcomes per encoder

Table S13 runs the certification pipeline of the main paper on each encoder's reduced operator ($k = 32$, equal dose). With the overfilled aperture (top; 24 phantoms $\times$ 3 noise seeds, as in the main paper) the no-plate control certifies 3 and 18 of 72 queries against 24 and 60 for the random plate, the in-focus DMD 19 and 36, and the trained plates 33 / 29 and 63. With a field-sized aperture (bottom; seed 0 only) the edge covers most of the 7.68 μm reduced field and gives the no-plate control depth structure.

	15 dB		25 dB	
encoder	cert. / ambig.	uncertain voxels	cert. / ambig.	uncertain voxels
random phase plate	24 / 48	7	60 / 12	0
trained plate, seed 0	33 / 39	4	63 / 9	0
trained plate, seed 1	29 / 43	4.5	63 / 9	0
defocused amplitude mask	24 / 48	10	59 / 13	0
no plate	3 / 69	54	18 / 54	15.5
in-focus DMD	19 / 53	17	36 / 36	4

	15 dB		25 dB	
encoder	cert. / ambig.	uncertain voxels	cert. / ambig.	uncertain voxels
phase plate	8 / 16	9.5	20 / 4	0
no plate	7 / 17	19	18 / 6	0
in-focus DMD	6 / 18	19	12 / 12	4
defocused amplitude mask	8 / 16	14	20 / 4	0

Table S13. Certified / ambiguous queries and median uncertain query voxels (of 64) per encoder on the reduced field. Top: overfilled aperture, 72 queries (24 phantoms $\times$ 3 seeds); the random plate reproduces the counts of the main paper exactly. Bottom: field-sized aperture, 24 queries (seed 0).

S6. Trained plate

Training. The plate loss (Eq. 5 of the main paper) is evaluated through the same simulation as the operator: angular-spectrum propagation over D, NA-0.8 vectorial relay, emission PSF at NA 0.1 and 8×8 pixel pooling, on the overfilled geometry (centre: the random plate of realisation 0, trained; surround: an independent draw, frozen, so the beam edges stay outside the field). Each Adam step (learning rate 0.05, 1000 steps) samples 24 lateral positions and 4 of the 256 directions (8 on the reduced field); the two seeds differ only in this sampling. Training changes the centre phase by 1.7–1.8 rad RMS and lowers the mean $|\rho|$ between depth columns of a training batch from 0.89 to 0.64 / 0.68 (full field, seeds 0 / 1) and from 0.88 to 0.78 / 0.82 (reduced field). It uses no specimen.

Certification and fabrication. Table S14 pairs every query of the trained plates with the same query of the random plate. Gains appear at both noise levels and no query is lost. The exact mean depth correlation of the reduced operator, $\langle|\rho|\rangle$ over all depth pairs and lateral positions, predicts the measured narrowing of the per-voxel bounds through the two-column rule of the main paper (predicted 0.80, measured 0.79 at 15 dB). Across encoders the same rule orders the measured widths correctly but overstates the differences where $\rho \to 1$: predicted / measured width relative to the random plate is 1.29 / 1.14 for the mask ($\langle|\rho|\rangle = 0.922$), 2.52 / 1.26 for the DMD (0.967) and 11.5 / 2.47 without the plate (0.999). Rounding the trained phase to 8 or 4 levels and adding uniform phase error to the 8-level plate raise the correlation slightly and keep 32–33 certificates.

Full size. Table S15 lists the full-field results behind the main paper. All encoders use the extended tuning grid of Sec. S4; the random plates select interior parameters at every

	15 dB			25 dB			
plate	cert.	+ / −	width	cert.	+ / −	width	$\langle\lvert\rho\rvert\rangle$
random plate	24	–	1.00	60	–	1.00	0.868
trained, seed 0	33	9 / 0	0.79	63	3 / 0	0.92	0.761
trained, seed 1	29	5 / 0	0.79	63	3 / 0	0.92	0.759
seed 0, 8 phase levels	32	8 / 0	0.80	63	3 / 0	0.92	0.767
seed 0, 4 phase levels	32	8 / 0	0.82	63	3 / 0	0.93	0.781
8 levels + 0.2 rad error	33	9 / 0	0.80	63	3 / 0	0.92	0.771
8 levels + 0.5 rad error	33	9 / 0	0.82	63	3 / 0	0.93	0.789

Table S14. Trained plate on the reduced field (72 queries per noise level). cert.: certified queries; + / −: queries certified only with this plate / only with the random plate; width: mean per-voxel bound width relative to the random plate; $\langle\lvert\rho\rvert\rangle$: exact mean depth correlation of the operator columns. No certificate contradicts the truth.

		$C=1$		$C=4$		$C=8$	
prior	encoder	PSNR	depth	PSNR	depth	PSNR	depth
pretrained PnP	random plate, realisation 0	35.73	0.357	35.35	0.384	34.81	0.431
	random plate, realisation 1	35.76	0.371	35.38	0.391	34.93	0.422
	random plate, realisation 2	35.80	0.369	35.45	0.392	34.85	0.438
	trained plate, seed 0	37.03	0.184	36.47	0.212	36.15	0.230
	trained plate, seed 1	37.12	0.173	36.49	0.206	36.21	0.234
TV	random plate, realisation 0	38.15	0.187	37.34	0.223	37.00	0.243
	trained plate, seed 0	39.29	0.135	38.18	0.166	38.61	0.153
	trained plate, seed 1	39.52	0.124	38.30	0.158	38.63	0.152
learned	random plate, realisation 0	39.18	0.161	38.29	0.184	37.11	0.221
	trained plate, seed 0	40.82	0.109	39.89	0.130	38.83	0.164

Table S15. Random and trained plates at full size with three priors (mean over 12 test crops; $C > 1$: five direction subsets per crop). Depth: local depth error in μm.

C. Paired over crops, each trained plate has the lower depth error than each random plate on 12 of 12 crops at every C, and the higher PSNR on 12 of 12 at $C = 1$ and on 9–12 of 12 at $C = 4$ and 8. With TV (the λ grid and edge rule of Tab. S12; every selection interior) both trained plates beat the random plate on 10–12 of 12 crops in PSNR and 11–12 of 12 in depth at every C. Their TV results are not monotone in C: at $C = 4$ validation selects $\lambda = 1$, which ties $\lambda = 0.3$ within 0.01 dB on the vascular validation crops but over-smooths the neurons.

S7. Learned prior and hybrid

Data and training. Training volumes come from SOM009 (cell 1, stacks 8 and 9), an animal that contributes no test cell; stack 10 is held out to monitor denoising. Planes whose mean intensity drops below half the median of their neighbours, or repeat the previous plane, are excluded. 459 training crops are cut on the $16 \times 48 \times 48$ reconstruction grid with the preprocessing of the test volumes. The 2D DnCNN (17 layers, 64 channels) is trained on random xy, xz and yz slices for 30k steps (batch 64), the 3D DnCNN (12 layers, 48 channels) on whole volumes for 15k steps (batch 8), both with Adam (10^{-3}), $\sigma \sim \mathcal{U}(0, 0.25)$, flips, lateral transposes and intensity scaling in $[0.3, 1]$; on held-out crops they gain 16.6 and 16.8 dB at $\sigma = 0.1$.

Hybrid. The hybrid adds to the learned prior an isotropic TV branch with weight λ_{TV} scaled from the standalone TV solver's selection (scales 0.3, 1, 3) and a box constraint, tuned with the extended grid. Before any test result we fixed that the reported solver per encoder and compression is the one with the best validation PSNR among TV, learned and hybrid. Validation selects the hybrid for the plate and TV for the DMD at every C (bold in Tab. S16). On the neuron test crops the hybrid matches the learned prior (higher PSNR on 4–7 of 12 crops) and beats TV on 8–11 of 12; the vascular validation crops favour the TV term more than the neurons do.

A second DMD pattern set. To check that the DMD's failure is not one unlucky draw, we built the DMD operator again with independent binary patterns (centre seed 101, surround 102; everything else unchanged) and reran pretrained PnP, TV and the learned prior with fresh tuning. Every solver reproduces the original pattern set to within 0.06 dB and 0.07 μm and stays at the depth error of a flat estimate (Tab. S17).

Where the prior's gain goes. For each lateral column of a crop we split the reconstruction error into its mean over depth (a wrong total intensity) and the zero-mean remainder (intensity in the wrong depth); the two squared parts add to the total. Table S18 gives the depth part's share and the learned / pretrained ratio of each part. With the plate and the mask the learned prior removes lateral and depth error, the latter on every crop; with the DMD it removes lateral error only.

		PSNR (dB)				depth error (μm)			
C	encoder	PnP	TV	learned	hybrid	PnP	TV	learned	hybrid
1	phase plate	35.73	38.15	39.18	**38.77**	0.357	0.187	0.161	**0.175**
	amp. mask	35.24	37.60	38.44	–	0.347	0.191	0.159	–
	no plate (4×)	30.87	32.78	31.33	–	0.753	0.477	0.632	–
	in-focus DMD	28.96	**28.48**	29.02	28.97	2.168	**2.281**	2.246	2.233
4	phase plate	35.35	37.34	38.29	**37.99**	0.384	0.223	0.184	**0.202**
	amp. mask	34.83	36.79	37.58	–	0.393	0.239	0.198	–
	no plate (4×)	30.40	32.09	30.71	–	1.036	0.706	0.914	–
	in-focus DMD	28.91	**28.49**	28.95	28.99	2.168	**2.327**	2.248	2.240
8	phase plate	34.81	37.00	37.11	**37.22**	0.431	0.243	0.221	**0.226**
	amp. mask	33.70	36.74	36.69	–	0.523	0.237	0.237	–
	no plate (4×)	30.18	31.81	30.58	–	1.102	0.737	0.922	–
	in-focus DMD	28.83	**28.53**	28.82	28.88	2.169	**2.311**	2.249	2.234

Table S16. Every encoder with every solver (mean over 12 test crops, realisation 0, equal dose). Bold: solver selected on validation among TV, learned and hybrid (hybrid run for plate and DMD only). No plate: four-times aperture (Sec. S1).

		$C=1$		$C=4$		$C=8$	
solver	patterns	PSNR	depth	PSNR	depth	PSNR	depth
pretrained PnP	set 0	28.96	2.168	28.91	2.168	28.83	2.169
	set 1	28.95	2.167	28.90	2.168	28.81	2.169
TV	set 0	28.48	2.281	28.49	2.327	28.53	2.311
	set 1	28.48	2.289	28.52	2.314	28.47	2.249
learned	set 0	29.02	2.246	28.95	2.248	28.82	2.249
	set 1	29.05	2.238	28.95	2.245	28.81	2.243

Table S17. In-focus DMD with the original pattern set (set 0) and an independent one (set 1); mean over 12 test crops, depth error in μm. Flat estimate: 2.148 μm.

		depth part of error		learned / pretrained		depth lower
encoder	C	pretrained	learned	lateral	depth	(crops)
phase plate	1	94 %	96 %	0.32	0.58	12 / 12
	8	93 %	95 %	0.48	0.71	12 / 12
amp. mask	1	93 %	96 %	0.32	0.61	12 / 12
	8	91 %	95 %	0.37	0.55	12 / 12
in-focus DMD	1	99 %	99 %	0.74	1.03	6 / 12
	8	99 %	99 %	0.77	1.04	4 / 12

Table S18. Lateral and depth parts of the squared reconstruction error (12 test crops; $C = 8$: draw 0). Ratios: median over crops.

S8. Detection: what can replace scan frames?

The main experiments detect at NA 0.1 on 12×12 pixels of 2.56 μm, so nearly all information comes from the 256 directions. We ask whether better detection can replace directions. Excitation is unchanged (random plate, realisation 0, overfilled aperture); only the emission NA (0.1 or 0.3) and the camera pooling (12×12, 24×24 or 48×48 pixels) change, and configurations are compared at a fixed number of measurements $M = k\times$pixels. Every configuration detects the same total number of photons per specimen as the reference, so the larger collection of NA 0.3 is not credited. Operators hold only the rows of the selected frames (three direction subsets for $k < 256$) and match the live forward model to $< 10^{-6}$; the learned prior is tuned per configuration on the validation crops with the extended grid (all picks interior).

Table S19 gives three results. At NA 0.1 finer pixels cannot replace directions: 16 frames on 48×48 pixels lose 1.5 dB and raise the depth error from 0.15 to 0.23 μm on every crop, because the NA-0.1 emission spot (Airy radius 3.2 μm) is wider than the pixels, which resample the same blur. At NA 0.3 (radius 1.1 μm) the trade works: 16 frames on 48×48 pixels or 64 on 24×24 reconstruct as well as all 256 frames at NA 0.3 and better than the NA-0.1 reference on 12 of 12 crops in PSNR and 8 or 11 of 12 in depth, a 4- to 16-fold shorter scan. With 8 times fewer measurements depth needs directions: two frames on 48×48 pixels give 0.49 μm against 0.22–0.23 μm with 8 or 32 frames. Detection is thus part of the optics that decide what the measurements allow; the camera can carry lateral detail, but depth still comes from the angle scan through the plate.

M	NA	pixels	pitch	frames	PSNR	FG-SSIM	depth	crops > ref.
36,864	0.1	12×12	2.56	256	39.17	0.914	0.154	–
	0.3	12×12	2.56	256	41.33	0.949	0.123	12 / 8
	0.1	48×48	0.64	16	37.69	0.888	0.230	0 / 0
	0.3	48×48	0.64	16	41.72	0.956	0.134	12 / 8
	0.3	24×24	1.28	64	41.82	0.954	0.115	12 / 11
4,608	0.1	12×12	2.56	32	37.02	0.872	0.227	0 / 0
	0.3	24×24	1.28	8	38.53	0.906	0.215	5 / 0
	0.3	48×48	0.64	2	35.64	0.839	0.494	0 / 0

Table S19. Detection NA and camera pooling against scan frames at fixed measurement count M and equal detected photons (random plate, learned prior; mean over 12 test crops, three direction subsets for fewer than 256 frames). Pitch in μm; depth: local depth error in μm; crops > ref.: crops with higher PSNR / lower depth error than the reference (first row). The reference is an independent run (noise draw and tuning) of the plate entry of Tab. S16.

S9. Formulation details

Variables and set. The reduced field has $N = 576$ voxels; the query region Q is the central $4 \times 4 \times 4$ block (64 voxels, 6-adjacency, 144 edges) and the remaining 512 voxels are free nuisance variables in $[0, 1]$. Each measurement row is rescaled by its largest coefficient before optimisation; all reported residuals are recomputed in unscaled units. For certificates the set is enlarged outward, $\mathcal{C}_\delta = \{\mathbf{x} : \ \ell_i - \delta\|a_i\|_1 \le a_i^\top \mathbf{x} \le h_i + \delta\|a_i\|_1\}$ with $\delta = 10^{-7}$, so that a certificate on $\mathcal{C}_\delta$ is robust to the solver's feasibility tolerance; witnesses must lie in the unenlarged $\mathcal{C}$ with residual at most 10^{-8} (scaled) and 10^{-6} (unscaled).

Minimum bottleneck (separator form). $b_{\min} = \min_{\mathbf{x}\in\mathcal{C}} \max_P \min_{v\in P} x_v$ is computed through vertex-separator duality. Binary s_v marks the side of v ($s_p = 1$, $s_q = 0$), binary z_v marks membership in the separator, and

$$\begin{aligned} \min\ t \ \text{ s.t. } & |s_u - s_v| \le z_u + z_v \quad \forall (u,v) \in E_Q, \\ & x_v \le t + (1 - t_{\text{lo}})(1 - z_v) \quad \forall v \in Q, \qquad \mathbf{x} \in \mathcal{C}. \end{aligned} \tag{S1}$$

Any feasible (s, z) defines a vertex set $S = \{v : z_v = 1\}$ that separates p from q, and $t \ge \max_{v\in S} x_v \ge b(\mathbf{x})$; conversely every separator is representable. The query is certified connected when the solver's dual bound on $\mathcal{C}_\delta$ exceeds $\tau + \eta$, $\eta = 10^{-6}$.

Maximum bottleneck (flow form). Binary w_v selects a path; a unit flow from p to q is supported on selected vertices ($0 \le f_{uv} \le w_u$, $f_{uv} \le w_v$, flow conservation), and $x_v \ge t - t_{\text{hi}}(1 - w_v)$; maximising t gives $b_{\max}$. The query is certified disconnected when the dual bound is below $\tau - \eta$.

Decision versions and witnesses. Fixing $t = \tau \pm \mu$ turns each program into a feasibility problem whose solution, if any, is a candidate witness $\mathbf{x}_C$ ($b \ge \tau + \mu$) or $\mathbf{x}_D$ ($b \le \tau - \mu$). Every candidate is re-checked independently: residual in unscaled units, active set at τ, connectivity by breadth-first search and by union–find, and threshold margin $\ge \eta$. Only re-checked pairs count as proven ambiguity.

Route order and guards. (i) Per-voxel envelope: 2×64 LPs on $\mathcal{C}_\delta$, warm-started, with failed solves retried cold and a feasible-point check; (ii) witness-first decision MIPs; (iii) LP path/separator enumeration when at most 16 voxels are uncertain and at most 200 induced paths exist; (iv) the joint MIP verifier. A certificate that contradicts a validated witness or a verified feasible point is reported as numerically undecided, and timeouts are never counted as ambiguity. All programs are solved with HiGHS [3] (single thread, tolerances 10^{-9}, zero relative MIP gap).

S10. Reduced operator and phantoms

The reduced operator uses the optics of the full system (532 nm, NA 0.8/0.1, one random plate with 0.32 μm features at $D = 6.6\,\mu$m, 256 directions, $2 \times 2 \times 2$ patch integration, equal dose) on a smaller field: optical grid 96×96 at 0.08 μm, analysis grid $8 \times 24 \times 24$ at 0.32 μm, reconstruction grid $4 \times 12 \times 12$ at 0.64 μm, and 3×3 camera pixels of 2.56 μm per pattern. It agrees with the live simulator to $\le 3.1 \times 10^{-7}$ relative ℓ_2 error on 10 volumes, 3 geometries and all 576 re-simulated columns, and $\mathsf{A} = \mathsf{BU}$ holds to 1.3×10^{-7}; negative controls (column order, axis swap, depth flip) give 8.4%, 7.4% and 3.5% error. All entries are positive; singular values range from 125.3 to 0.058 at $k = 256$ (condition number 2.2×10^3), and the nested direction subsets have condition numbers 2.0×10^2, 5.6×10^2, 8.9×10^5 and 2.9×10^3 at $k = 16, 32, 64, 128$. The cosine between operator columns of depth-adjacent voxels is 0.958, 0.869 and 0.802 at separations of 1, 2 and 3 planes. Table S20 lists the 24 phantoms.

id	family	structure (query region, $4 \times 4 \times 4$)	p	q
P01	connected	straight filament through all depths	(0,5,5)	(3,5,5)
P02	connected	3D staircase filament, U(0.5,1)	(0,4,4)	(3,7,7)
P03	connected	lateral staircase filament in one plane	(1,4,4)	(1,7,7)
P04	disconnected	two separated filaments (z=0 and z=3)	(0,4,4)	(3,7,7)
P05	disconnected	filament with one missing voxel	(0,5,4)	(3,6,7)
P06	disconnected	filament with one weak (0.3) bridge voxel	(0,6,4)	(3,5,7)
P07	connected	two alternative-depth routes (z=0 and via z=3), both bright	(0,4,5)	(0,7,5)
P08	connected	alternative-depth routes, shallow route dim (0.15-0.25)	(0,4,5)	(0,7,5)
P09	disconnected	laterally crossing filaments at different depths	(0,4,5)	(3,6,7)
P10	connected	crossing filaments joined by a vertical connector	(0,4,5)	(3,6,7)
P11	disconnected	four isolated beads (endpoints on beads)	(0,4,4)	(3,7,7)
P12	connected	bead chain (adjacent beads of varying brightness)	(0,7,4)	(3,7,7)
P13	connected	three alternative routes, intensities 0.5-0.8	(0,4,4)	(1,4,7)
P14	connected	P01 filament + U(0,0.15) background over the field	(0,5,5)	(3,5,5)
P15	disconnected	P04 + bright distractor blobs outside the ROI	(0,4,4)	(3,7,7)
P16	connected	graded filament $1.0 \rightarrow 0.45$	(0,7,4)	(3,4,7)
P17	connected	thick (2-voxel) vertical filament	(0,5,5)	(3,6,5)
P18	connected	in-plane filament at z=0 (mirror-depth control: mirror plane empty)	(0,4,6)	(0,7,6)
P19	connected	ring in plane z=1, endpoints opposite	(1,4,4)	(1,7,7)
P20	connected	dim filament (0.42-0.48), just above tau = 0.4	(0,4,4)	(3,7,7)
P21	connected	random lattice walk (9 voxels)	(0,4,4)	(1,6,5)
P22	unknown	two random walks (connectivity decided by truth BFS)	(0,7,7)	(3,4,4)
P23	unknown	sparse random volume (15% of ROI voxels U(0.3,1)); fixed corners	(0,4,4)	(3,7,7)
P24	unknown	dense random volume U(0,1) over the whole field	(0,4,4)	(3,7,7)

Table S20. The 24 phantoms. Coordinates are global (z, a, b); the query region is $z \in [0, 3]$, $a, b \in [4, 7]$. For the "unknown" families, connectivity is decided by the truth.

S11. Reconstruction versus ambiguity: all cases

All 26 reconstructions (two estimators, 13 cases) lie inside $\mathcal{C}(\mathbf{y})$ (Tab. S21 and Fig. S7). In the dense phantom P24 the reconstruction fits the data more closely than the truth (largest normalised residual 0.24 vs. 0.83 at 15 dB, 0.001 vs. 0.83 at 25 dB) while reporting the wrong topology. The total variation of the opposite-topology witnesses is close to that of the truth for the filament phantoms (P01: truth 11.6, witnesses 13.2 and 12.6; P20: 17.5, 17.7 and 20.4; P06: 23.4, 24.3 and 23.0), so these volumes are not rough artefacts. The TV weight of the reconstruction baseline was selected on ten calibration phantoms: $\lambda = 3 \times 10^{-4}$ at 15 dB (mean calibration PSNR 46.9 dB vs. 46.2 dB without TV) and $\lambda = 0$ at 25 dB.



Figure S7. All 13 cases in the layout of main-paper Fig. 4. Certified cases show no opposite-topology volume because none exists in $\mathcal{C}(\mathbf{y})$.

case	dB	truth	recon. TV	PSNR	recon. LS	consistent C	consistent D	data decide
P01	15	C 0.60 / 0.61	C 0.64 / 0.62	51.5	C 0.62 / 0.62	C 0.49 / 1.00	D 0.40 / 1.00	ambiguous
P06	15	D 0.30 / 0.72	D 0.30 / 0.68	46.4	D 0.30 / 0.69	C 0.40 / 1.00	D 0.40 / 1.00	ambiguous
P07	15	C 0.72 / 0.83	C 0.59 / 0.83	41.5	C 0.55 / 0.83	C 0.41 / 1.00	D 0.40 / 1.00	ambiguous
P13	15	C 0.54 / 0.79	C 0.45 / 0.74	31.2	C 0.42 / 0.74	C 0.40 / 1.00	D 0.20 / 1.00	ambiguous
P20	15	C 0.42 / 0.54	C 0.42 / 0.55	48.1	C 0.41 / 0.56	C 0.45 / 1.00	D 0.10 / 1.00	ambiguous
P24	15	C 0.43 / 0.83	D 0.00 / 0.23	7.8	D 0.00 / 0.24	C 0.40 / 1.00	D 0.00 / 1.00	ambiguous
P13	25	C 0.54 / 0.79	C 0.53 / 0.74	50.3	C 0.53 / 0.74	C 0.56 / 1.00	D 0.40 / 1.00	ambiguous
P20	25	C 0.42 / 0.54	C 0.42 / 0.56	68.2	C 0.42 / 0.56	C 0.41 / 1.00	D 0.40 / 1.00	ambiguous
P24	25	C 0.43 / 0.83	D 0.20 / 0.00	13.6	D 0.20 / 0.00	C 0.40 / 1.00	D 0.40 / 1.00	ambiguous
P09	15	D 0.00 / 0.61	D 0.01 / 0.64	50.5	D 0.00 / 0.64	–	–	cert. D
P18	15	C 0.77 / 0.80	C 0.75 / 0.72	58.2	C 0.76 / 0.70	–	–	cert. C
P06	25	D 0.30 / 0.72	D 0.30 / 0.69	65.3	D 0.30 / 0.69	–	–	cert. D
P07	25	C 0.72 / 0.83	C 0.70 / 0.83	59.1	C 0.70 / 0.83	–	–	cert. C

Table S21. All 13 reduced-field cases. Each entry: topology at $\tau = 0.4$ (C/D), bottleneck b, and the largest normalised residual $\max_i |(\mathbf{A}\mathbf{x} - \mathbf{y})_i|/z\sigma$ (inside $\mathcal{C}$ iff ≤ 1). Reconstruction PSNR is against the coarse truth.

S12. Ambiguity versus directions and photon budget

Figure S8 and Tab. S22 report, for the queries that are proven ambiguous with $k = 32$ directions, how many become certified with k directions (row confidence set, $\tau = 0.4$, three seeds pooled, after the 600 s rerun). The decision rule (primary metric: certified fraction at $k = 256$ with fixed total photons; outcomes A ≥ 0.30, B 0.10–0.30, C < 0.10) was fixed before any aggregate was computed; the outcome is C.

Numerical validation. 1776 problems were solved with no errors and no wrong certificates. The 110 problems that hit the 60 s limit were rerun at 600 s with nothing else changed: 59 became proven ambiguities, 5 became certificates and 1 numerical boundary case remained. The 1325 stored witness volumes were re-validated on independently rebuilt problems (maximum unscaled residual 9.6×10^{-10}), and all 18 certificates not issued by the envelope were re-derived by the other exact route. Total compute: 696 CPU-hours.

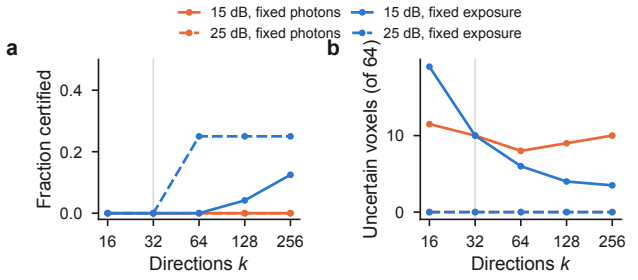


Figure S8. (a) Fraction of the queries proven ambiguous at $k = 32$ that become certified with k directions, with fixed total photons or fixed per-frame exposure ($8\times$ photons at $k = 256$). (b) Median uncertain query voxels over all 72 queries. Three seeds; no query remained undecided.

k	16	32	64	128	256
total dose / baseline	1×	1×	1×	1×	1×
certified	0	0	0	0	0
proven ambiguous	48	48	48	48	48
undecided	0	0	0	0	0
median uncertain voxels	11.5	10	8	9	10
mean bound width	0.338	0.306	0.291	0.301	0.320

(a) 15 dB, fixed total photons ($n = 48$)

k	16	32	64	128	256
total dose / baseline	1×	1×	1×	1×	1×
certified	0	0	0	0	0
proven ambiguous	12	12	12	12	12
undecided	0	0	0	0	0
median uncertain voxels	0	0	0	0	0
mean bound width	0.132	0.114	0.099	0.095	0.096

(b) 25 dB, fixed total photons ($n = 12$)

k	16	32	64	128	256
total dose / baseline	0.5×	1×	2×	4×	8×
certified	0	0	0	2	6
proven ambiguous	48	48	48	46	42
undecided	0	0	0	0	0
median uncertain voxels	19	10	6	4	3.5
mean bound width	0.392	0.306	0.245	0.211	0.184

(c) 15 dB, fixed per-frame exposure ($n = 48$)

k	16	32	64	128	256
total dose / baseline	0.5×	1×	2×	4×	8×
certified	0	0	3	3	3
proven ambiguous	12	12	9	9	9
undecided	0	0	0	0	0
median uncertain voxels	0	0	0	0	0
mean bound width	0.149	0.114	0.087	0.073	0.065

(d) 25 dB, fixed per-frame exposure ($n = 12$)

Table S22. Outcomes per number of directions k. "Median uncertain voxels" and "mean bound width" describe the per-voxel envelope over all 72 queries.

set	dB	photons	ambig. k=32	certified k=256	ambiguous	undecided	truth $\notin \mathcal{C}$
row+SVD	15	8×	16	4	11	1	3
row+SVD	15	fixed	16	0	15	1	3
row+SVD	25	8×	4	1	3	0	3
row+SVD	25	fixed	4	0	4	0	3
Poisson (Garwood)	15	8×	17	2	15	0	1
Poisson (Garwood)	15	fixed	17	0	17	0	1
Poisson (Garwood)	25	8×	5	2	3	0	0
Poisson (Garwood)	25	fixed	5	0	5	0	0

Table S23. Secondary confidence sets (seed 0, $k \in \{32, 256\}$). Row+SVD adds 576 calibrated projections of the 2304 rows; Poisson uses Garwood intervals at the same family level.

S13. Total-variation prior

$\mathcal{C}_R(\mathbf{y}) = \{\mathbf{x} \in \mathcal{C}(\mathbf{y}) : \sum_{(u,v)\in E} |x_u - x_v| \le R\}$ over the 1488 6-neighbour edges of the reduced field is encoded exactly with one auxiliary variable $e_{uv} \ge |x_u - x_v|$ per edge and $\sum e_{uv} \le R$. $R_{\mathrm{cal}} = 31.9$ is the largest TV among ten calibration phantoms and was fixed before any test case. The certification pipeline (600 s per solve) was run on the nine ambiguous cases of Sec. S11 at R_{cal}, $2R_{\mathrm{cal}}$ and $4R_{\mathrm{cal}}$ (Tab. S24). In all 18 runs whose budget the truth satisfies, two validated opposite-topology witnesses exist inside $\mathcal{C}_R$, and they are as regular as the truth (P20 at 25 dB: TV 17.3 and 17.5 vs. 17.5; P01 at 15 dB: 12.2 and 12.9 vs. 11.6). No run produced a certificate. Only TV was tested; a neurite-specific learned prior was not.

case	dB	TV(truth)	R_{cal}	$2R_{\mathrm{cal}}$	$4R_{\mathrm{cal}}$
P01	15	11.6	ambiguous	ambiguous	ambiguous
P06	15	23.4	ambiguous	ambiguous	ambiguous
P07	15	33.9	ambiguous†	ambiguous	ambiguous
P13	15	44.3	empty set†	ambiguous	ambiguous
P20	15	17.5	ambiguous	ambiguous	ambiguous
P24	15	454.0	ambiguous†	ambiguous†	ambiguous†
P13	25	44.3	empty set†	ambiguous	ambiguous
P20	25	17.5	ambiguous	ambiguous	ambiguous
P24	25	454.0	empty set†	empty set†	empty set†

Table S24. Outcomes under a TV budget. †: the true object violates the budget. "Empty set": no volume satisfies both the data and the prior.

S14. Fine-grid model mismatch

Fine objects live on the $8 \times 24 \times 24$ analysis grid: *sub-voxel occupancy* (each voxel's value carried by 4 of its 8 sub-voxels at twice the value, so $\mathsf{R}\mathbf{x}_f$ equals the coarse truth)

fine structure	dB	noise only	+ mismatch (max)	+ mismatch (99.9%)
sub-voxel	15	0/24	21/24	21/24
sub-voxel	25	0/24	21/24	21/24
multiplicative	15	1/24	22/24	22/24
multiplicative	25	0/24	22/24	20/24

Table S25. Held-out coverage of the coarse truth (8 phantoms $\times$ 3 seeds). All failures with the allowance come from phantom P18 (a filament confined to the outermost depth plane), whose residual reaches 25%, outside the calibration range.

and a *multiplicative* model $\mathbf{x}_f = \mathrm{clip}(\mathsf{U}\mathbf{x} \cdot U(0,2), 0, 1)$. Measurements are $\mathsf{B}\mathbf{x}_f$ from the fine-grid simulator plus the Gaussian noise of the main study. The allowance $\epsilon_i = c(|y_i| + z\sigma)$ uses c calibrated on ten calibration phantoms $\times$ three seeds: $c = 0.78$ / 0.58 (sub-voxel; maximum / 99.9% row quantile) and 0.37 / 0.25 (multiplicative). Calibration residuals $\|\mathsf{B}\mathbf{x}_f - \mathsf{A}\mathsf{R}\mathbf{x}_f\|/\|\mathsf{B}\mathbf{x}_f\|$ were 5.8–13.0% and 2.3–7.1%.

fine structure	dB	model	n	truth $\in \mathcal{C}$	empty	certified	ambiguous	undecided	contradicts truth
multiplicative	15	noise	8	0	7	0	1	0	0
multiplicative	15	noise + allowance (max)	8	8	0	0	8	0	0
multiplicative	15	noise + allowance (99.9%)	8	8	0	0	8	0	0
multiplicative	25	noise	8	0	7	0	1	0	0
multiplicative	25	noise + allowance (max)	8	8	0	0	8	0	0
multiplicative	25	noise + allowance (99.9%)	8	7	0	0	8	0	0
multiplicative	exact	exact	8	0	7	0	1	0	0
sub-voxel	15	noise	8	0	7	0	1	0	0
sub-voxel	15	noise + allowance (max)	8	7	0	0	8	0	0
sub-voxel	15	noise + allowance (99.9%)	8	7	0	0	8	0	0
sub-voxel	25	noise	8	0	7	0	1	0	0
sub-voxel	25	noise + allowance (max)	8	7	0	0	8	0	0
sub-voxel	25	noise + allowance (99.9%)	8	7	0	1	7	0	1
sub-voxel	exact	exact	8	0	8	0	0	0	0

Table S26. Query outcomes under model mismatch (seed 0). "Contradicts truth" counts certificates opposite to the true connectivity regardless of coverage; the single case (P18, sub-voxel, 25 dB, 99.9% rule) occurred when the allowance did not cover the truth.

S15. Preliminary single-stack study

Before the encoder comparison, the phase-plate system was evaluated on eight prespecified crops of *one* neuron stack with a Gaussian relative-noise model at 15 dB and the field-sized aperture (Tab. S27); the main paper uses an independent crop set from six cells. On a held-out crop of that study, voxel integration lowers the reconstruction error throughout the iterations relative to a coarse-grid simulation (Fig. S9), and foreground SSIM is 0.913, 0.832 and 0.761 at $C = 1$, 16 and 32 (depth errors 0.177, 0.354 and $0.443\ \mu$m); the higher compressions are exploratory and not part of the test-set evaluation.

C	depth err. (μm)	FG-SSIM	Δdepth (μm)	ΔPSNR (dB)
1	0.497 ± 0.183	0.679 ± 0.238	–	–
4	0.550 ± 0.170	0.652 ± 0.252	$+0.053 \pm 0.016$	-0.45 ± 0.35
8	0.626 ± 0.207	0.588 ± 0.273	$+0.129 \pm 0.032$	-0.91 ± 0.45

Table S27. Preliminary study: eight crops of one stack, phase plate, mean $\pm$ s.d. over crops of five-subset averages; Δ: paired change against each crop's own $C = 1$ result.

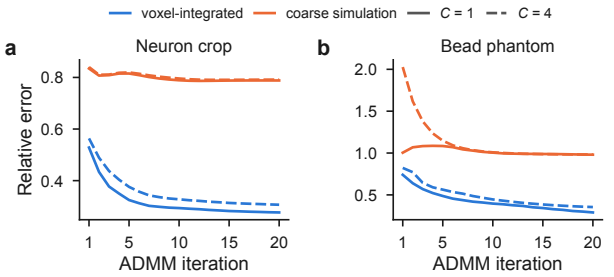


Figure S9. Relative reconstruction error over ADMM iterations with the voxel-integrated and the coarse-simulation operator, on (a) a neuron crop and (b) a bead phantom of the preliminary study (mean of three noise realisations).

S16. Implementation details

Optical simulation, operator assembly and reconstruction run in PyTorch and JAX on NVIDIA RTX 6000 Ada and RTX 4500 Ada GPUs; PnP and TV reconstruction use SCICO [1] with a pretrained 17-layer DnCNN [5]. A full-size operator is assembled in 0.6–7 minutes; a PnP reconstruction takes 2–9 s and a TV reconstruction 9–150 s on one GPU. Certification uses HiGHS [3] on CPUs with BLAS threads fixed to one. All tuned quantities of the structural study (TV weight, TV budget, mismatch allowance) were selected on

ten calibration phantoms (P02, P03, P04, P05, P08, P10, P12, P16, P17, P21) and fixed before any test case was run.

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